

DATA-DRIVEN CASE STUDY



INTERNET SHUTDOWNS IN INDIA

Balancing Public Safety and Digital Rights

This case study examines 920 internet shutdown events recorded in India between 2016 and 2025. It uses the Access Now Shutdown Tracker Optimization Project (STOP) dataset as its primary source, supported by the SFLC.in India tracker, to analyze how shutdowns vary by year, location, duration, reported cause, network type, and restriction type.

The findings are presented through a repeatable Python and SQL workflow and an interactive Power BI dashboard. The analysis is descriptive: it documents patterns in the recorded data and keeps the limits of that data visible.

CASE STUDY TITLE	Case Study: Internet Shutdowns in India, 2016–2025
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Executive Summary

This case study examines **920 internet shutdown events** recorded in India between 2016 and 2025. The Government of India does not maintain a central record of shutdown orders issued by states and Union Territories. The study therefore uses the Access Now Shutdown Tracker Optimization Project (STOP) dataset as its primary source, supported by the SFLC.in India tracker for context and verification.

The analysis describes how recorded shutdowns vary by year, location, duration, reported cause, network type, and restriction type. A repeatable Python pipeline prepares the data, and an interactive Power BI dashboard presents the results. The main findings are:

- **Frequency.** Recorded shutdowns peaked at 134 in 2018, stayed above 100 a year through 2021, and fell to 63 in 2025.
- **Location.** Jammu and Kashmir has the most shutdown-state relationships (488), nearly six times the next highest state, Rajasthan (83).
- **Duration.** The median shutdown lasted 2 days, but the average was about 6.6 days because a small number of very long shutdowns, up to 551 days, raise it.
- **Reported causes.** Political instability, protests, and communal violence account for 605 of the 920 records (about 66%).
- **Networks and restrictions.** Mobile connectivity was affected in 890 of the 920 records (about 97%), and 871 records (about 95%) are classified as shutdowns rather than throttling.

The analysis is descriptive. It documents patterns in the recorded data and does not establish why shutdowns were ordered or whether they achieved their intended outcomes.

WHY INTERNET SHUTDOWNS MATTER

Internet access is now essential for communication, education, business, banking, healthcare, news, and public services. At the same time, authorities in India sometimes suspend internet services during protests, communal violence, security concerns, examinations, elections, and other events.

A GAP IN OFFICIAL RECORDS

No single publicly available government dataset combines all shutdown orders issued by Indian states and Union Territories. In a written reply to the Lok Sabha in March 2025, the government stated that the Department of Telecommunications does not centrally maintain the details of these orders, because police and public order are State subjects (The Week, 2025).

DATA SOURCES USED IN THIS STUDY

Because of this gap, researchers and civil-society organizations have built their own datasets to track shutdown events. This case study uses the **Access Now STOP dataset** as its main source. Access Now documents shutdown cases using technical measurements and contextual information, such as news reports and personal accounts, and the current dataset covers 2016–2025 (Access Now, n.d.). The **SFLC.in India tracker** serves as a supporting source for Indian context and verification. It records shutdown events in India using publicly available reports, government orders, and other sources (SFLC.in, n.d.).

Statement of the Problem

THE PROBLEM

Authorities may use internet shutdowns when they judge that limiting communication is necessary for public safety or to control unrest. However, a shutdown also interrupts everyday communication and access to digital services.

The central problem is that shutdown information is scattered across different sources and is not maintained centrally by the Indian government. This makes it difficult to compare shutdowns by year, location, duration, reason, and network type. This project organizes publicly available shutdown data and applies data analysis to identify patterns that can help researchers and decision-makers understand how shutdowns are used across India.

MAIN RESEARCH QUESTION

What patterns can be observed in internet shutdowns in India in terms of frequency, duration, location, reported reason, and network type?

SUPPORTING QUESTIONS

1. How has the number of recorded shutdowns changed from year to year?
2. Which states and Union Territories have the highest number of recorded shutdown events?
3. How long do recorded shutdowns typically last?
4. What are the most common reported causes?
5. Which types of networks are most often affected?
6. How do shutdown characteristics differ across time and location?

OBJECTIVES

- Analyze year-wise internet shutdown trends in India.
- Study the geographic distribution of recorded shutdowns.
- Measure shutdown duration using the available records.
- Analyze the reported causes of shutdowns.
- Compare the affected network types.
- Build an automated data-cleaning and analysis pipeline.
- Present the findings in an interactive Power BI dashboard.

Decision Criteria and Assumptions

DATA SOURCES

Primary dataset: Access Now STOP. The Shutdown Tracker Optimization Project covers documented shutdown events from 2016 through 2025. Its fields include dates, locations, shutdown type, affected networks, causes, government justification, legal information, and other event details. Access Now describes STOP as a way to document and contextualize shutdowns, built from technical and contextual sources.

Supporting source: SFLC.in. The Internet Shutdowns Tracker is an India-specific record of shutdown events. It provides location, nature, affected services, descriptions, and links to supporting government orders or other sources where available.

GUIDING PRINCIPLES AND ASSUMPTIONS

1. **Descriptive, not causal.** The analysis identifies patterns. It does not try to prove that one factor directly caused another.
2. **Recorded, not verified.** Cause categories reflect what the source dataset records. They are not treated as independently verified explanations.
3. **Relationships, not unique events, at state level.** Events that affected several states are counted once for each state involved.
4. **No guessed values.** Invalid or missing values are flagged and excluded from calculations instead of being replaced with estimates.
5. **A main source, not a census.** The STOP dataset is used as the main public record, with SFLC.in as a supporting source, but it is not assumed to capture every shutdown.

Data Analysis

The cleaned dataset contains **920 recorded shutdown events** from 2016 to 2025. The findings that follow are descriptive. They show what the recorded data contain, not a complete census of every shutdown.

920

Recorded shutdown events, 2016–2025

2 days

Median duration

6.6 days

Average duration (approx.)

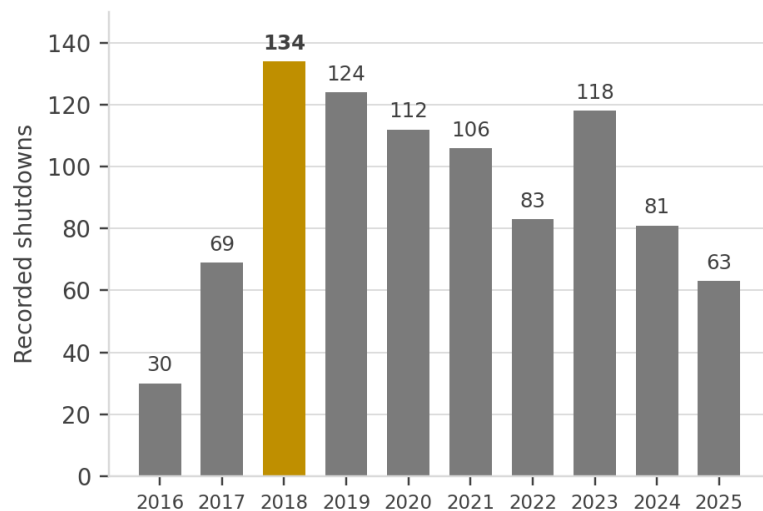
551 days

Longest recorded shutdown

DURATION

For records with usable duration information, the median duration is 2 days and the average is about 6.6 days. The maximum recorded duration is 551 days. The large gap between the median and the average shows that a small number of very long shutdowns strongly affect the average. For this reason, the dashboard displays both measures instead of relying on the average alone.

DATA ANALYSIS Year-Wise Pattern



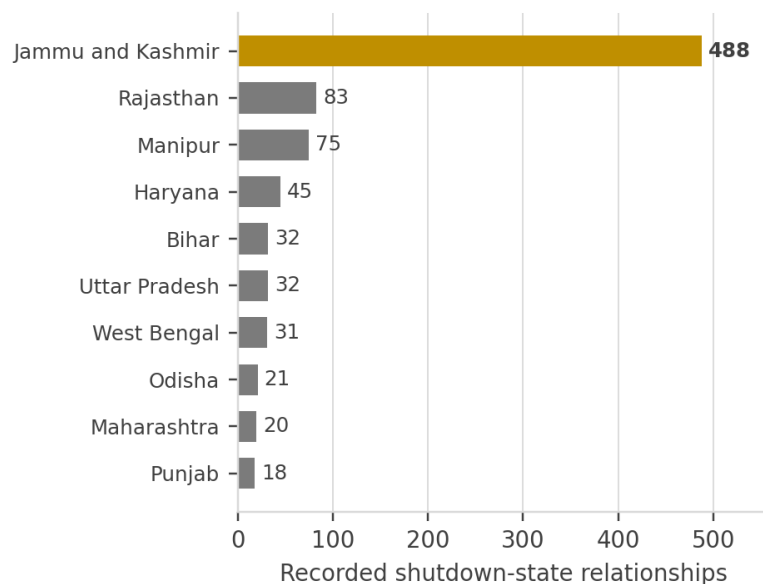
2018 has the highest number of recorded events (134), more than four times the 30 events recorded in 2016.

The pattern is not a simple continuous increase. The count rises through 2018, remains relatively high through 2021, falls in 2022, rises again in 2023, and then declines in 2024 and 2025.

DATA NOTE

Access Now's 2025 report states that it documented **65** shutdowns in India during 2025 (Access Now, 2026), while the extracted dataset contains **63**. The final project should document the exact STOP dataset version used and investigate this difference before treating the 2025 figure as final.

DATA ANALYSIS Geographic Distribution



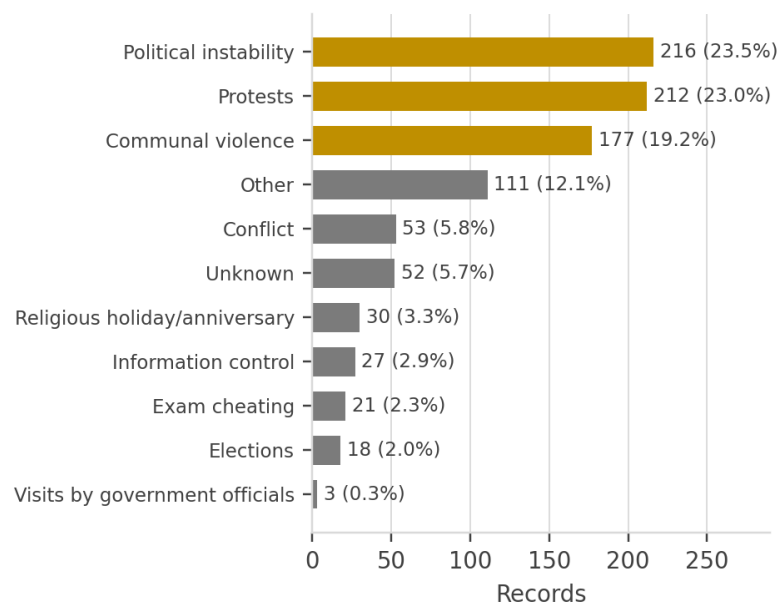
Some shutdowns affected more than one state. The state-level figures therefore count **shutdown-state relationships**, not unique shutdown events.

Jammu and Kashmir has the largest number by a wide margin (488), followed by Rajasthan (83), Manipur (75), and Haryana (45). Bihar and Uttar Pradesh follow with 32 each.

HOW TO READ THESE FIGURES

These numbers describe recorded relationships in the dataset. They should not be read as a measure of the severity or importance of shutdowns in any state.

DATA ANALYSIS Reported Causes



The dataset provides a structured `actual_cause` field with eleven categories.

Political instability (216), protests (212), and communal violence (177) are the three most frequently recorded causes. Together they account for 605 of the 920 records, or about 66%.

HOW TO READ THESE FIGURES

These categories describe the recorded cause in the source dataset. They should not automatically be interpreted as independently verified explanations for why a shutdown was necessary or effective.

DATA ANALYSIS Networks and Restrictions

AFFECTED NETWORK

Network	Records	Share
Mobile	571	62.1%
Broadband, Mobile	319	34.7%
Unknown	28	3.0%
Broadband	2	0.2%

RESTRICTION TYPE

Restriction type	Records	Share
Shutdown	871	94.7%
Shutdown + Throttle	28	3.0%
Throttle	19	2.1%
Unknown	2	0.2%

Mobile connectivity is the dominant network category: 571 records list mobile alone, and another 319 list broadband and mobile together. Most records are classified as full shutdowns (871). The restriction-type field also allows the analysis to separate full shutdowns from throttling and from records that contain both.

Limitations

This case study has several important limitations. The table below sets out each one and how the study responds to it.

Limitation	What it means	How the study responds
1. Not a government census	The Government of India has stated that the Department of Telecommunications does not centrally maintain state and Union Territory suspension orders.	Findings are described as recorded patterns, not as official totals.
2. Recorded events may not cover every event	Access Now aims to build a comprehensive dataset but relies on technical measurement and contextual information, such as news reports and personal accounts.	SFLC.in is used as a supporting source, and the exact STOP dataset version is to be documented.
3. State extraction is imperfect	The source <code>area_name</code> field contains cities, districts, regions, and multi-state descriptions, so some records need careful geographic interpretation.	Multi-state events are retained as such, and state counts are labeled as relationships.
4. Duration is incomplete	Some records do not contain enough information to calculate a duration.	Invalid values are flagged and excluded, not guessed, and both median and average are reported.
5. Cause is not proof of effectiveness	A recorded cause or government justification does not show that a shutdown achieved its intended public-safety outcome.	Causes are described as recorded categories, not as verified explanations.

Alternative Courses of Action

Several analytical choices shaped this study. For each one, the table compares a plausible alternative with the course adopted and explains the reasoning.

Decision	Alternative approach	Course adopted	Reasoning
Multi-state events	Assign each event to a single state.	Retain multi-state events and count shutdown-state relationships.	Avoids forcing records into one state and misrepresenting geography.
Invalid durations	Replace negative or missing values with estimates.	Flag invalid values and exclude them from duration calculations.	Keeps guessed values out of the analysis.
Duration measure	Report the average only.	Report both the median and the average.	The median is less affected by a few very long shutdowns (up to 551 days).
Data source	Look for a single official government dataset.	Use STOP as the primary source and SFLC.in as a supporting source.	No central government dataset exists, and STOP covers 2016–2025 with structured fields.
Data preparation	Clean the data manually each time.	Build an automated Python pipeline.	Makes preparation repeatable when a new dataset version is released.
Analytical framing	Test whether one factor caused another.	Describe patterns in the recorded data.	The data support description, not causal claims.

Recommendations

Based on the analysis, we recommend the following practices for anyone who uses this dataset and dashboard to inform research or policy discussion.

- 1. Report median and average duration together.** A few very long shutdowns raise the average well above the typical case.
- 2. Describe state-level counts as recorded relationships.** They are not a measure of the severity or importance of shutdowns in each state.
- 3. Treat causes as recorded, not verified.** A recorded cause or justification is not evidence that a shutdown was necessary or effective.
- 4. Document the dataset version and reconcile the 2025 count.** Record the exact STOP version used and investigate the difference between 63 and 65 shutdowns before treating the 2025 figure as final.
- 5. Cross-check against SFLC.in.** Use the tracker, and the government orders and other sources it links to, to verify individual records where possible.
- 6. Refresh the analysis with the automated pipeline.** Rerun the workflow whenever a new version of the source dataset becomes available.

Implementation Plan

FROM RAW DATA TO DASHBOARD

The project follows a single workflow, from the source dataset to the final dashboard. The analysis focuses on descriptive patterns.



ROLE OF AUTOMATION

A Python pipeline makes data preparation repeatable. It automatically:

- Loads the dataset
- Filters records for India
- Cleans dates
- Extracts states
- Handles duration values
- Runs data-quality checks
- Creates analysis-ready CSV files

The goal is to avoid repeating manual cleaning each time a new version of the source dataset becomes available.

DASHBOARD

The Power BI dashboard is a single-page decision-support view. It combines four key indicators (total recorded shutdowns, median duration, number of affected states and Union Territories, and longest recorded shutdown) with six main visuals and five filters.

Users can filter by year, state or Union Territory, cause, network, and restriction type. The full specification is provided in Exhibit B.

Conclusion

This case study used public shutdown data to examine how internet shutdowns in India vary across time, geography, duration, reported cause, and network type. The analysis shows that recorded shutdowns are not evenly distributed across years or locations, and that their duration varies widely. It also shows that mobile networks are affected in nearly all recorded events, and that political instability, protests, and communal violence are among the most frequently recorded causes.

The results offer a structured way to examine documented shutdowns while keeping the limitations of the underlying data visible. The accompanying Power BI dashboard lets users explore these patterns by year, location, cause, network, and restriction type.

Exhibits and Schedules

EXHIBIT A: DATA PREPARATION PROCEDURE

Step	Action
1. India filtering	The Access Now dataset initially contains shutdown records from many countries. Only records where country = India were retained.
2. Column selection	Relevant fields were kept: date, location, cause, duration, shutdown type, network affected, government justification, legal information, and event description.
3. Date cleaning	Start and end dates were converted to a standard date format. Start year, start month, and month name fields were created.
4. State extraction	State and Union Territory names were extracted from the area_name field. Records that affected more than one state were retained as multi-state events instead of being forced into a single state.
5. Duration cleaning	The original duration field was retained. Invalid negative duration values were flagged and excluded from usable duration calculations instead of being replaced with guessed values.

EXHIBIT B: POWER BI DASHBOARD SPECIFICATION

Key indicators	Main visuals	Filters
- Total recorded shutdowns - Median duration - Number of affected states and Union Territories - Longest recorded shutdown	- Shutdowns by year - Shutdowns by state or Union Territory - Duration distribution - Reported causes - Affected network - Restriction type	- Year - State or Union Territory - Cause - Network - Restriction type

References

References follow APA (7th edition) style.

Access Now. (n.d.). #KeepItOn Shutdown Tracker Optimization Project (STOP). <https://www.accessnow.org/shutdown-tracker/>

Access Now. (2026, March 31). *One region, half the world's shutdowns: Internet shutdowns in the Asia Pacific in 2025* [Press release]. <https://www.accessnow.org/press-release/one-region-half-the-worlds-shutdowns-internet-shutdowns-in-the-asia-pacific-in-2025/>

SFLC.in. (n.d.). *Internet shutdown tracker*. <https://internetshutdowns.in/>

The Week. (2025, March 13). *Impact of internet shutdowns not assessed by Telecommunications dept*. <https://www.theweek.in/news/india/2025/03/13/impact-of-internet-shutdowns-not-assessed-by-telecommunications-dept-pemmasani-chandra-sekhar-said.html>